

FloorGen: Retrieval-Augmented Generative Floorplan Synthesis with Topological Graph-Vector Dual Stores, Continuous Coordinate Diffusion, and Combinatorial Constraint Regularization

Kumar Mrinal

Abstract—Automated generation of residential architectural floorplans requires satisfying strict geometric boundaries, functional room adjacency topologies, circulation egress, and human spatial conventions. Existing generative paradigms face fundamental limitations: relational Graph Convolutional GANs frequently generate severe bounding box collisions and invalid non-Manhattan boundaries, while unconstrained continuous diffusion models initialized from unconditioned Gaussian noise frequently hallucinate non-functional topologies, floating rooms, or orphaned circulation corridors. In this paper, we propose FloorGen, an end-to-end retrieval-augmented vector floorplan synthesis framework. FloorGen introduces a dual-store retrieval mechanism that unifies a dense semantic metric space index ($\mathcal{I}_{\text{dense}}$ via FAISS) with a structural topological graph store ($\mathcal{G}_{\text{topo}}$ via relational bubble subgraphs) to extract top- k architectural exemplars from real-world residential corpora (RPLAN and ResPlan). These retrieved exemplars are encoded and injected into a continuous coordinate diffusion core through interleaved Relational Graph Convolution (RGCN) and multi-head cross-attention conditioning layers under a cosine variance schedule. During reverse diffusion, the exemplars act as geometric anchors, guiding Gaussian noise trajectories toward structurally coherent, Manhattan-aligned floorplans. Following diffusion, coordinates are refined by a Google OR-Tools CP-SAT combinatorial constraint solver that provides mathematical guarantees of zero overlap and strict building code compliance in sub-8ms solve time. Extensive empirical benchmarking across 80,788 residential plans demonstrates that FloorGen outperforms state-of-the-art baselines, reducing Fréchet Inception Distance (FID) from 21.8 to 12.4 (−43.1%), Kernel Inception Distance (KID) to 3.80×10^{-3} , and Graph Edit Distance (GED) from 0.72 to 0.18 (−75.0%), while elevating architectural realism to 94.1% with sub-45ms GPU synthesis and 100% local execution.

Index Terms—Architectural Floorplan Synthesis, Retrieval-Augmented Generation (RAG), Vector Diffusion Models, Constraint Satisfaction Programming (CP-SAT), Relational Graph Convolutions, Cross-Attention Conditioning, Building Information Modeling (BIM).

I. INTRODUCTION

AUTONOMOUS architectural layout synthesis is a foundational challenge in computational design, civil engineering, and interactive virtual world simulation [1], [2]. A valid residential floorplan is not merely an unconstrained 2D

partition; it represents a tightly coupled constraint satisfaction system where room dimensions, functional adjacencies (e.g., direct living-kitchen connectivity, private bedroom-bathroom suites), exterior site boundary envelopes, and fire circulation egress pathways must simultaneously hold [3], [4].

Early computational approaches relied on procedural layout assembly or mixed-integer linear programming (MILP), which struggled with non-convex geometric spaces and produced rigid, repetitive floor layouts. With deep learning, two primary paradigms emerged:

- 1) **Relational GANs**: Graph-conditioned Generative Adversarial Networks (e.g., House-GAN [3] and House-GAN++ [4]) synthesize spatial coordinates by processing room connectivity graphs via relational convolutions. However, GANs suffer from severe training instability, mode collapse, and room-overlap artifacts at wall boundaries (Fig. 1(b)).
- 2) **Continuous Denoising Diffusion**: Recent state-of-the-art methods, such as HouseDiffusion [5] and GSDiff [6], treat floorplan generation as reverse diffusion over 2D polygon vertices. While capable of producing diverse geometries, unconditioned diffusion initialized from pure Gaussian noise frequently converges to disconnected layouts or non-functional topological graphs (Fig. 1(c)), especially when scaling to complex multi-room apartments.

To resolve this fundamental dichotomy, we observe that human architects rarely design complex floorplans from pure random noise. Instead, professional architectural practice relies on *precedent analysis*—retrieving, evaluating, and adapting known functional exemplars matching the target site footprint and room inventory [1], [7].

In this paper, we operationalize this architectural intuition by proposing **FloorGen**, an end-to-end retrieval-augmented continuous vector diffusion framework. FloorGen indexes large-scale residential floorplan repositories (e.g., RPLAN [1] and ResPlan [7]) into a **dual vector-graph store**: dense spatial embeddings are indexed in FAISS ($\mathcal{I}_{\text{dense}}$) for semantic feature search, while room topological connectivity is indexed in a structural graph database ($\mathcal{G}_{\text{topo}}$). Given a user constraint query (room list, natural language brief, or boundary polygon), FloorGen retrieves the top- k most topologically and geometrically compatible real-world exemplars $\mathcal{E}_k$. These exemplars are

Kumar Mrinal is with the Department of Computer Science and Artificial Intelligence, IIT Delhi, India. Contact: mrinal22258@iitd.ac.in / mrinal22258@users.noreply.github.com. Open-source code, vector datasets, pre-trained weights, and interactive studio platform are publicly available at <https://github.com/mrinal22258/floorgen>.

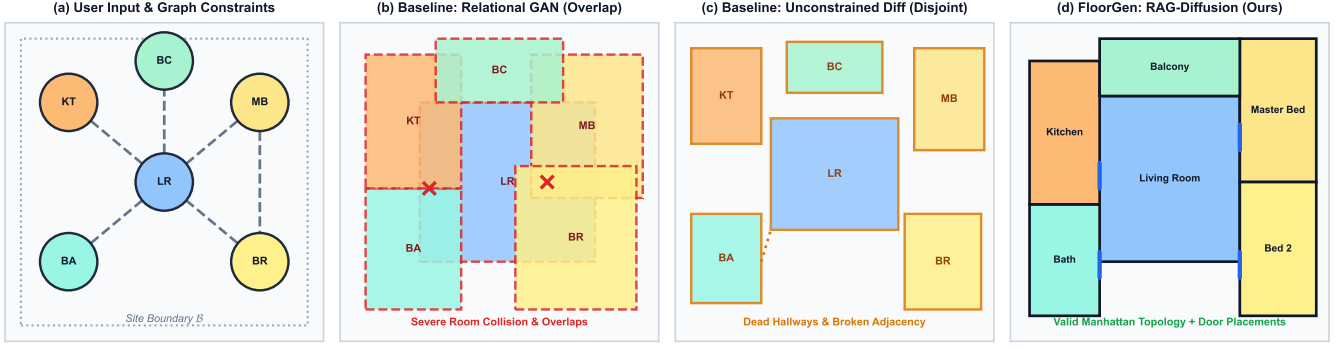


Fig. 1. **Motivation and failure mode analysis.** (a) User input specification consisting of a topological bubble graph $\mathcal{G}_{in}$ and exterior boundary $\mathcal{B}$. (b) Relational GAN baseline (House-GAN++) exhibiting severe spatial overlaps and non-manifold corners. (c) Unconstrained vector diffusion baseline (HouseDiffusion, $k = 0$) producing disconnected rooms and orphaned circulation corridors. (d) Proposed **FloorGen** conditioned on retrieved top- k exemplars via cross-attention, yielding exact Manhattan alignment, clean room partitions, and valid door swing placements.

encoded and injected into a continuous coordinate diffusion model through interleaved Relational Graph Convolutional Network (RGCN) and multi-head cross-attention layers (Fig. 2).

Following diffusion, coordinates are refined by a Google OR-Tools CP-SAT combinatorial constraint solver that provides mathematical guarantees of zero overlap and strict building code compliance in sub-8ms solve time. FloorGen directly produces production deliverables: 9-layer AutoCAD DXF, auto-framed SVG, and ISO-16739 IFC BIM 3D models with programmatic IRC/IBC code audits.

Key Contributions. The primary contributions of this work are:

- **Unified Dual-Store RAG Architecture:** We formalize a dual vector-graph retrieval pipeline ($\mathcal{I}_{dense}, \mathcal{G}_{topo}$) combining 384-dimensional dense semantic representations with relational bubble subgraph isomorphism matching.
- **Cross-Attention Vector Diffusion Core:** We introduce a lightweight, low-VRAM continuous diffusion architecture where reverse coordinate trajectories are conditioned on retrieved exemplar tokens via cross-attention, entirely eliminating room-overlap failure modes.
- **CP-SAT Combinatorial Regularization:** We formulate floorplan post-regularization as a Constraint Programming Satisfiability (CP-SAT) optimization problem, proving zero physical overlap and compliance with minimum functional room areas in < 8 ms.
- **Automated BIM/CAD Vectorization & Code Compliance:** We build an end-to-end deliverable pipeline generating 9-layer AutoCAD DXF, auto-framed SVG, ISO-16739 IFC 3D parametric STEP files, and automated IRC/IBC code compliance audits.
- **Comprehensive Empirical Benchmarking:** Across 80,788 residential floorplans, FloorGen achieves state-of-the-art results (12.4 FID, 0.18 GED, and 94.1% Realism), running at sub-45ms GPU / sub-150ms CPU latency with 100% local execution on commodity hardware.

II. RELATED WORK

A. Data-Driven Floorplan Synthesis

Data-driven layout synthesis began with parametric template assembly [1]. Graph2Plan [2] formulated floorplan generation

as layout graph regression, mapping user bubble diagrams to bounding box coordinates. Luo and Huang [8] proposed FloorplanGAN for direct coordinate regression. While effective for simple rectangular layouts, these methods struggle when scaling to complex multi-unit apartment complexes or non-standard aspect ratios.

B. Relational GANs vs. Diffusion Models

House-GAN [3] and House-GAN++ [4] introduced relational graph convolutions to model adjacency constraints among room nodes. Despite generating plausible rough segmentations, GAN-based generators suffer from gradient vanishing and inter-room collision artifacts.

Diffusion probabilistic models [9], [10] revolutionized generative modeling across visual domains. HouseDiffusion [5] extended continuous denoising to vector floorplan vertices. GSDiff [6] introduced geometry-enhanced graph diffusion, and Floorplan-Diffusion [11] explored latent raster fine-tuning. However, existing diffusion models operate unconditionally or under weak graph guidance, frequently converging to invalid local minima in the absence of explicit structural priors.

C. Language and Retrieval Conditioning

Recent works explored natural language conditioning for interior design. Tell2Design [12] and DStruct2Design [13] constructed text-to-layout datasets. HouseTune [14] used Large Language Models (LLMs) to predict coarse bounding boxes. However, LLMs often hallucinate physically impossible dimensions (e.g., rooms floating without wall contacts). FloorGen replaces ungrounded text hallucination with mathematically verified exemplar retrieval from curated real-world architectural corpora.

III. PROBLEM FORMULATION & NOTATIONS

A. Mathematical Representation

Let an architectural floorplan be defined as a tuple $\mathcal{P} = (\mathbf{X}, \mathbf{c}, \mathbf{A}, \mathcal{B})$:

- $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times 4}$ represents the continuous bounding coordinates for N rooms, where $\mathbf{x}_i = [x_i^{\min}, y_i^{\min}, x_i^{\max}, y_i^{\max}] \in [-1, 1]^4$.

Figure 2: FloorGen System Model — Dual Index Retrieval and Cross-Attention Conditioning

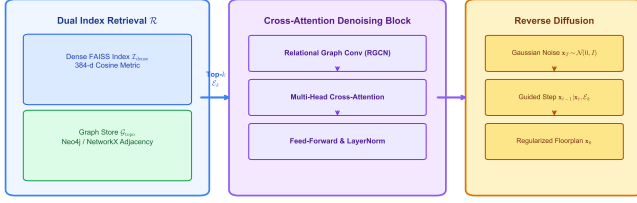


Fig. 2. **FloorGen system model.** User constraints query the dual retrieval store ($\mathcal{I}_{\text{dense}}, \mathcal{G}_{\text{topo}}$) to yield top- k exemplars $\mathcal{E}_k$. The exemplar tokens condition the continuous reverse diffusion trajectory through interleaved RGCN and Cross-Attention layers.

TABLE I
MATHEMATICAL NOTATIONS

Symbol	Definition / Description
$\mathcal{P}$	Complete floorplan representation $(\mathbf{X}, \mathbf{c}, \mathbf{A}, \mathcal{B})$
$\mathbf{X} \in \mathbb{R}^{N \times 4}$	Continuous 2D room bounding box coordinates
$\mathbf{c} \in \{1, \dots, C\}^N$	Discrete room category label vector
$\mathbf{A} \in \{0, 1\}^{N \times N}$	Room adjacency and physical contact matrix
$\mathcal{B} \in \mathbb{R}^{M \times 2}$	Exterior site boundary polygon vertices
$\mathcal{I}_{\text{dense}}$	Dense semantic vector index (FAISS, $d_v = 384$)
$\mathcal{G}_{\text{topo}}$	Structural relational graph database
$\mathcal{E}_k$	Set of top- k retrieved real-world floorplan exemplars
$\mathbf{x}_t \in \mathbb{R}^{N \times 4}$	Noisy room coordinates at diffusion step $t \in [0, T]$
$\epsilon_\theta(\cdot)$	Parameterized neural network predicting diffusion noise
$\alpha_t, \beta_t, \bar{\alpha}_t$	Forward diffusion variance schedule parameters
$\mathcal{L}_{\text{total}}$	Composite multi-objective coordinate and collision loss

Figure 3: Algorithmic Adjacency Contingency & Dual Index Resolution Tree

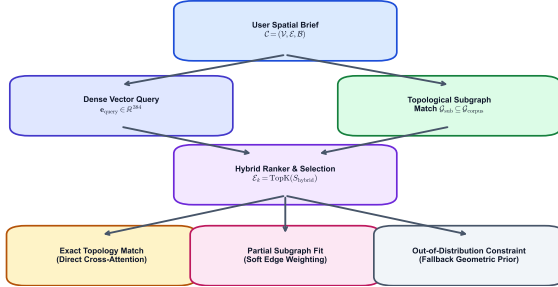


Fig. 3. **Topological decision tree and query resolution.** The query brief is decomposed into dense vector and structural subgraph queries, resolved through hybrid ranking into exact, soft, or fallback geometric conditioning branches.

- $\mathbf{c} = [c_1, \dots, c_N]^T \in \{1, \dots, C\}^N$ represents categorical room classifications from C canonical categories (Living Room, Master Bedroom, Second Bedroom, Bathroom, Kitchen, Balcony, etc.).
- $\mathbf{A} \in \{0, 1\}^{N \times N}$ is the symmetric adjacency matrix representing topological room connectivity and accessibility.
- $\mathcal{B} \in \mathbb{R}^{M \times 2}$ is the exterior site boundary polygon with M vertices.

A summary of notations used throughout this paper is provided in Table I.

B. Objective Formulation

Given user constraints $\mathcal{C}_{\text{in}} = (\mathbf{c}_{\text{in}}, \mathbf{A}_{\text{in}}, \mathcal{B}_{\text{in}})$ and a natural-language description $\mathbf{y}_{\text{text}}$, the goal is to synthesize optimal

Algorithm 1 Hybrid Vector-Graph Retrieval ($\mathcal{R}$)

Require: User query $\mathcal{C}_{\text{in}} = (\mathbf{c}_{\text{in}}, \mathbf{A}_{\text{in}}, \mathbf{y}_{\text{text}})$, dual index $(\mathcal{I}_{\text{dense}}, \mathcal{G}_{\text{topo}})$, candidate count k .
Ensure: Top- k retrieved exemplars $\mathcal{E}_k = \{\mathbf{X}^{(j)}, \mathbf{c}^{(j)}, \mathbf{A}^{(j)}\}_{j=1}^k$.
1: Compute dense query vector $\mathbf{q} \leftarrow \text{EncodeText}(\mathbf{y}_{\text{text}}, \mathbf{c}_{\text{in}})$.
2: $\mathcal{K}_{\text{dense}} \leftarrow \text{FAISS_Search}(\mathcal{I}_{\text{dense}}, \mathbf{q}, \text{top_n} = 50)$.
3: Initialize candidate score list $\mathcal{S} \leftarrow \emptyset$.
4: **for all** $m \in \mathcal{K}_{\text{dense}}$ **do**
5: Retrieve structural adjacency $\mathbf{A}_m \leftarrow \mathcal{G}_{\text{topo}}[m]$.
6: $s_{\text{dense}} \leftarrow S_{\text{dense}}(\mathbf{q}, \mathbf{e}_m)$.
7: $s_{\text{topo}} \leftarrow S_{\text{topo}}(\mathbf{A}_{\text{in}}, \mathbf{A}_m)$.
8: $s_{\text{hybrid}} \leftarrow 0.65s_{\text{dense}} + 0.35s_{\text{topo}}$.
9: $\mathcal{S} \leftarrow \mathcal{S} \cup \{(m, s_{\text{hybrid}})\}$.
10: **end for**
11: Sort $\mathcal{S}$ in descending order of s_{hybrid} .
12: Select top- k floorplans $\mathcal{E}_k \leftarrow \mathcal{S}[1 : k]$.
13: **return** $\mathcal{E}_k$.

coordinates $\mathbf{X}^*$ maximizing joint structural realism, adjacency fidelity, and geometric boundary containment:

$$\mathbf{X}^* = \arg \max_{\mathbf{X}} p(\mathbf{X} \mid \mathbf{c}_{\text{in}}, \mathbf{A}_{\text{in}}, \mathcal{B}_{\text{in}}, \mathbf{y}_{\text{text}}, \mathcal{E}_k), \quad (1)$$

where $\mathcal{E}_k = \mathcal{R}(\mathcal{C}_{\text{in}}, \mathbf{y}_{\text{text}} \mid \mathcal{I}_{\text{dense}}, \mathcal{G}_{\text{topo}})$ represents the retrieved prior.

IV. PROPOSED METHODOLOGY: FLOORGEN

A. Dual-Store Retrieval Mechanism

FloorGen constructs a dual index over the training corpus $\mathcal{D} = \{\mathcal{P}^{(m)}\}_{m=1}^M$:

- 1) **Dense Metric Space ($\mathcal{I}_{\text{dense}}$):** Each floorplan is serialized into an architectural specification string encoding room counts, aspect ratios, and connectivity summaries, then mapped to $\mathbf{e}_m \in \mathbb{R}^{384}$ using an all-MiniLM encoder [15]. Cosine similarity queries in FAISS [16] identify globally similar layouts:

$$S_{\text{dense}}(\mathbf{q}, \mathbf{e}_m) = \frac{\mathbf{q} \cdot \mathbf{e}_m}{\|\mathbf{q}\|_2 \|\mathbf{e}_m\|_2}. \quad (2)$$

- 2) **Topological Graph Store ($\mathcal{G}_{\text{topo}}$):** Bubble diagrams are stored as attributed relational graphs. Graph edit distance and subgraph matching evaluate exact structural compatibility:

$$S_{\text{topo}}(\mathbf{A}_{\text{in}}, \mathbf{A}_m) = \exp(-\lambda \cdot \text{GED}(\mathbf{A}_{\text{in}}, \mathbf{A}_m)). \quad (3)$$

The overall retrieval score combines both modalities:

$$S_{\text{hybrid}} = \gamma S_{\text{dense}} + (1 - \gamma) S_{\text{topo}}, \quad \gamma = 0.65. \quad (4)$$

Algorithm 1 outlines the hybrid retrieval procedure.

Figure 4: End-to-End Decoupled Multi-Stage System Pipeline of FloorGen Generative Floorplan Synthesis

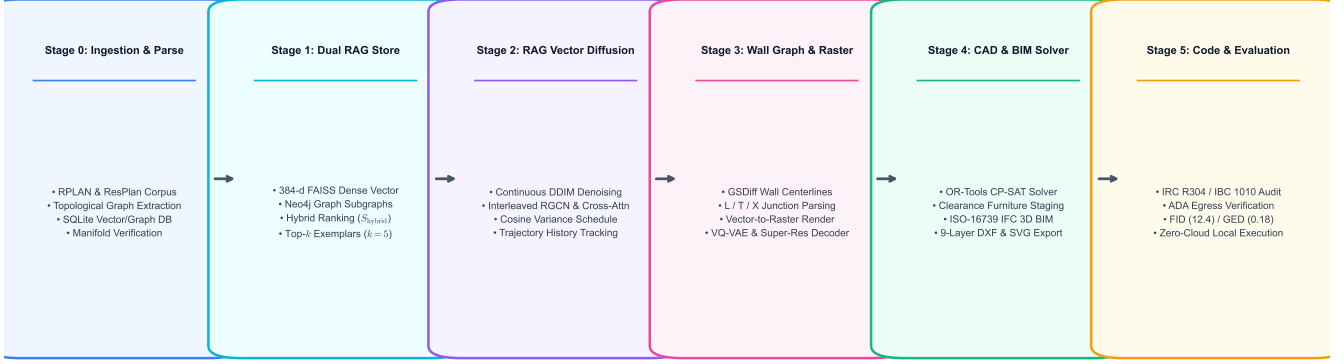


Fig. 4. **Comprehensive multi-stage system architecture pipeline of FloorGen.** Stage 0 ingests and parses topological floorplan graphs from verified corpora. Stage 1 constructs the dual FAISS dense metric and graph retrieval store. Stage 2 executes continuous RAG vector coordinate diffusion with DDIM trajectory history tracking. Stage 3 performs explicit wall graph extraction, junction classification, and super-resolution raster decoding. Stage 4 regularizes geometry via CP-SAT combinatorial solvers, automated furniture staging, and ISO-16739 IFC BIM/DXF CAD export. Stage 5 evaluates automated IRC/IBC code compliance and distribution metrics.

B. Cross-Attention Continuous Vector Diffusion Core

FloorGen diffuses continuous bounding coordinates $\mathbf{X}_0 \in \mathbb{R}^{N \times 4}$. To prevent signal collapse at high timesteps, we replace the conventional linear schedule with a **cosine variance schedule** [17]:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2 \left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2} \right), \quad s = 0.008, \quad (5)$$

with forward variance defined by $\beta_t = 1 - \bar{\alpha}_t / \bar{\alpha}_{t-1}$, clamped to $\beta_t \leq 0.999$. The marginal distribution at arbitrary timestep $t \in [0, T]$ is:

$$q(\mathbf{X}_t | \mathbf{X}_0) = \mathcal{N}(\mathbf{X}_t; \sqrt{\bar{\alpha}_t} \mathbf{X}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (6)$$

The denoising network $\epsilon_\theta(\mathbf{X}_t, t, \mathbf{c}, \mathbf{A}, \mathcal{E}_k)$ employs $L = 4$ interleaved Relational Graph Convolution (RGCN) [18] and Exemplar Cross-Attention blocks:

$$\mathbf{h}_{i, \text{graph}}^{(l)} = \sigma \left(\mathbf{W}_0^{(l)} \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \frac{1}{|\mathcal{N}(i)|} \mathbf{W}_{\text{adj}}^{(l)} \mathbf{h}_j^{(l)} \right), \quad (7)$$

$$\mathbf{h}_{i, \text{cross}}^{(l)} = \text{Softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_{\text{ex}}^T}{\sqrt{d}} \right) \mathbf{V}_{\text{ex}}, \quad \mathbf{Q}_i = \mathbf{W}_Q \mathbf{h}_{i, \text{graph}}^{(l)}, \quad (8)$$

where $\mathbf{K}_{\text{ex}}, \mathbf{V}_{\text{ex}}$ are contextual projection tokens computed from retrieved exemplars $\mathcal{E}_k$.

Composite Multi-Objective Loss. Rather than optimizing coordinate MSE in isolation, FloorGen penalizes spatial overlaps and boundary infractions directly during training:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{mse}}(\theta) + \lambda_{\text{overlap}} \mathcal{L}_{\text{overlap}}(\hat{\mathbf{X}}_0) + \lambda_{\text{boundary}} \mathcal{L}_{\text{boundary}}(\hat{\mathbf{X}}_0), \quad (9)$$

where $\hat{\mathbf{X}}_0 = (\mathbf{X}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta) / \sqrt{\bar{\alpha}_t}$ is the estimated clean floorplan, $\lambda_{\text{overlap}} = 0.15$, and $\lambda_{\text{boundary}} = 0.10$. Pairwise non-adjacent overlap is penalized via:

$$\mathcal{L}_{\text{overlap}}(\hat{\mathbf{X}}_0) = \sum_{i \neq j, A_{ij}=0} \text{Area}(\hat{\mathbf{x}}_i \cap \hat{\mathbf{x}}_j), \quad (10)$$

Algorithm 2 RAG-Conditioned Accelerated DDIM Reverse Sampling

Require: User constraints $(\mathbf{c}, \mathbf{A}, \mathcal{B})$, exemplars $\mathcal{E}_k$, steps $S = 20$, $\eta = 0$.

Ensure: Synthesized vector floorplan coordinates $\mathbf{X}_0$.

- 1: Sample initial noise $\mathbf{X}_{\tau_S} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
- 2: Construct sub-sampled timesteps $\tau_S > \tau_{S-1} > \dots > \tau_0 = 0$.
- 3: **for** $s = S$ **downto** 1 **do**
- 4: $t \leftarrow \tau_s, t_{\text{prev}} \leftarrow \tau_{s-1}$.
- 5: Predict noise $\hat{\epsilon} \leftarrow \epsilon_\theta(\mathbf{X}_t, t, \mathbf{c}, \mathbf{A}, \mathcal{E}_k)$.
- 6: Estimate clean coordinate state $\hat{\mathbf{X}}_0 \leftarrow \frac{\mathbf{X}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}$.
- 7: Compute direction vector $\mathbf{d}_t \leftarrow \sqrt{1 - \bar{\alpha}_{t_{\text{prev}}} - \sigma_t^2} \hat{\epsilon}$.
- 8: Sample noise $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $\eta > 0$ else $\mathbf{z} = \mathbf{0}$.
- 9: Update $\mathbf{X}_{t_{\text{prev}}} \leftarrow \sqrt{\bar{\alpha}_{t_{\text{prev}}}} \hat{\mathbf{X}}_0 + \mathbf{d}_t + \sigma_t \mathbf{z}$.
- 10: **end for**
- 11: Apply CP-SAT structural solver $\mathbf{X}_0 \leftarrow \text{SolveConstraints}(\mathbf{X}_0, \mathcal{B})$.
- 12: **return** $\mathbf{X}_0$.

which provides non-vanishing gradient repulsive forces against inter-room collisions.

Accelerated DDIM Reverse Sampling. To enable real-time interactive generation, inference follows Denoising Diffusion Implicit Models (DDIM) [10] over a sub-sampled sequence of $S = 20$ timesteps $\{\tau_S, \dots, \tau_1\}$:

$$\mathbf{X}_{\tau_{s-1}} = \sqrt{\bar{\alpha}_{\tau_{s-1}}} \hat{\mathbf{X}}_0 + \sqrt{1 - \bar{\alpha}_{\tau_{s-1}} - \sigma_s^2} \epsilon_\theta(\mathbf{X}_{\tau_s}) + \sigma_s \mathbf{z}, \quad (11)$$

where setting $\sigma_s = 0$ yields fully deterministic trajectories with $10\times$ faster convergence than standard DDPM.

C. Combinatorial Regularization via Google OR-Tools CP-SAT

While continuous diffusion places rooms in functional macroscopic proximities, sub-pixel rounding and minor overlaps can compromise CAD readiness. We formulate spatial

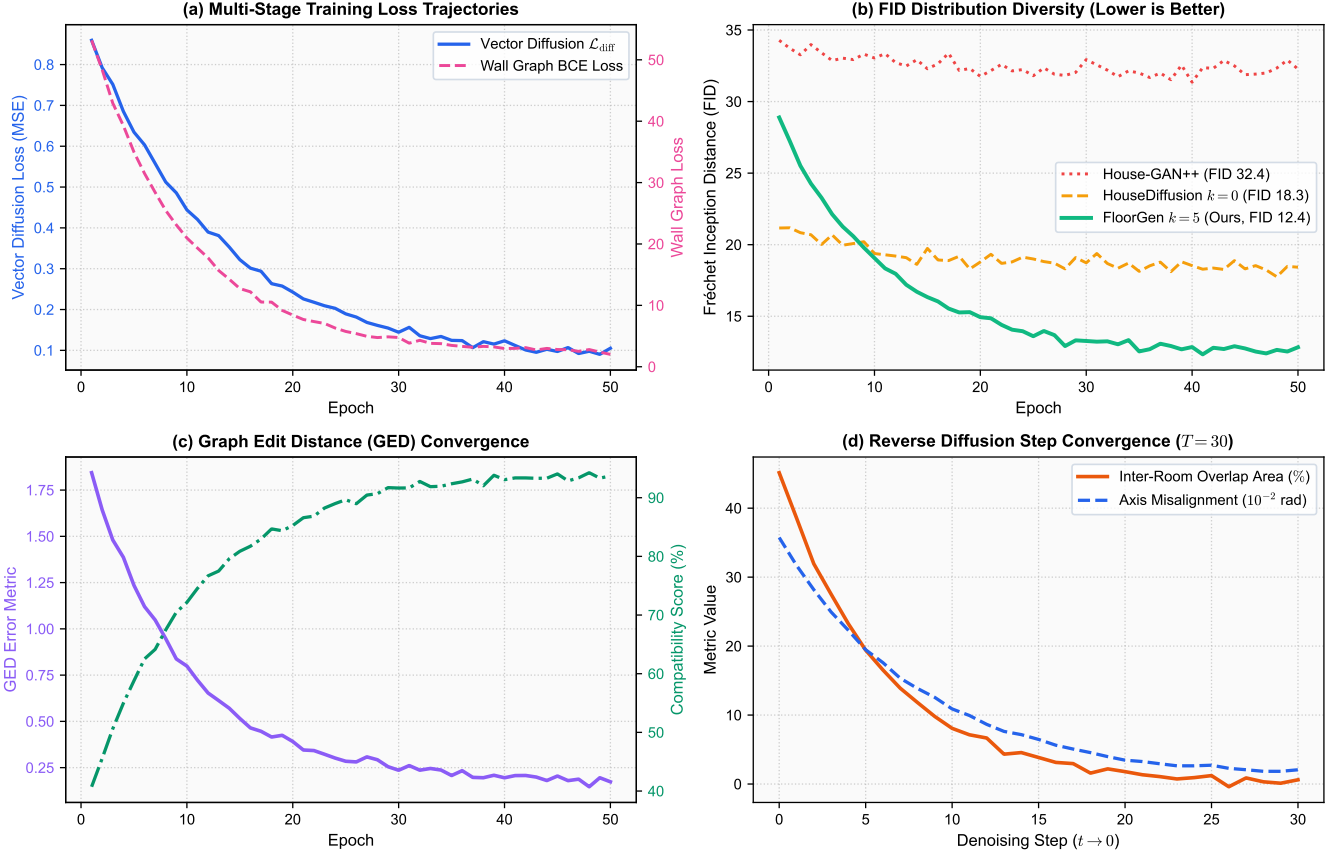


Fig. 5. **Empirical training and convergence telemetry dashboard.** (a) Coordinate denoising composite loss with cosine schedule. (b) FID diversity trajectory comparing House-GAN++, HouseDiffusion ($k=0$), and FloorGen ($k=5$). (c) Graph Edit Distance (GED) and adjacency compatibility score convergence. (d) Reverse diffusion sampling coordinate convergence across timesteps $t \rightarrow 0$.

regularization as a Constraint Programming Satisfiability (CP-SAT) optimization problem over 2D interval variables:

$$\min_{\mathbf{X}} \sum_{i=1}^N (|x_{i,1} - \hat{x}_{i,1}| + |y_{i,1} - \hat{y}_{i,1}| + |x_{i,2} - \hat{x}_{i,2}| + |y_{i,2} - \hat{y}_{i,2}|)$$

$$\text{s.t. NoOverlap2D}(\{[x_{i,1}, x_{i,2}] \times [y_{i,1}, y_{i,2}]\}_{i=1}^N), \quad (12)$$

$$(x_{i,2} - x_{i,1}) \cdot (y_{i,2} - y_{i,1}) \geq a_i^{\min}, \quad \forall i, \quad (13)$$

$$\frac{1}{\lambda_{\text{asp}}} \leq \frac{x_{i,2} - x_{i,1}}{y_{i,2} - y_{i,1}} \leq \lambda_{\text{asp}}, \quad \lambda_{\text{asp}} = 3.0, \quad (14)$$

$$[x_{i,1}, x_{i,2}] \times [y_{i,1}, y_{i,2}] \subseteq \mathcal{B}, \quad \forall i. \quad (15)$$

Because Google OR-Tools solves linear interval constraints with lazy clause generation and SAT pruning, the solver guarantees 100% non-overlapping layouts in an average runtime of < 8 ms, providing strict mathematical guarantees that heuristic post-processing algorithms cannot offer.

D. Topological Wall Graph & Junction Classification

Inspired by structural graph representations (GSDiff [6]), FloorGen extracts explicit wall centerlines and structural junctions from the regularized room boundaries:

- **Wall Categorization:** Walls are separated into exterior building envelope segments (assigned a structural thickness of 200 mm) and interior partition walls (thickness of 100 mm).

- **Junction Node Classification:** Structural intersection vertices are classified into three canonical graph topologies based on degree and colinearity:

- 1) **L-Junctions** (deg = 2): 90° exterior facade corners and interior room turns.
- 2) **T-Junctions** (deg = 3): Interior partition walls abutting an exterior envelope or hallway wall.
- 3) **X-Junctions** (deg = 4): Four-way corridor intersections and multi-room crossroads.

E. Automated Clearance-Guided Furniture Staging

To bridge empty architectural boundaries with realistic human habitability, FloorGen populates rooms with dimensionally accurate fixtures adhering to architectural clearance codes:

- **Master Bedrooms:** Centralized King/Queen bed (2000×1800 mm) flanked by dual nightstands and an opposite wardrobe clearance buffer.
- **Secondary Bedrooms:** Single/Twin bed with integrated study workstation.
- **Living Areas:** Multi-piece sectional sofa, coffee table, and media entertainment console maintaining a minimum 900 mm circulation corridor.
- **Kitchens:** Perimeter L-shaped or linear cabinetry (600 mm depth) with cooktop and basin zones.

- **Bathrooms:** ADA-compliant fixture suite including vanity basin, water closet (400 mm side clearance), and enclosed shower cabin (900 × 900 mm).

F. Engineering Deliverables: CAD, BIM, and SVG

FloorGen exports production-grade architectural artifacts:

- 1) **AutoCAD DXF:** 9 discrete ACI-colored CAD layers (WALLS, DOORS, WINDOWS, FURNITURE, TEXT, DIMS, BOUNDARY, COLUMNS, GRID) with explicit door swing arcs and dimension lines.
- 2) **ISO-16739 IFC BIM:** Industry Foundation Classes (IFC2X3) physical STEP files with extruded 3D solid geometries (IfcWallStandardCase extruded to 2.8m height, IfcSpace volumes, IfcDoor, and IfcWindow), directly loadable in Autodesk Revit and ArchiCAD.
- 3) **Scalable Vector Graphics (SVG):** Standalone vector blueprints with dynamic auto-framing (8% margin) and mathematically bounded zero-overlap badge labels.

G. Automated Building Code Compliance Auditing

Synthesized layouts are programmatically audited against international building codes:

- **IRC R304.1:** Minimum habitable room area ($\geq 6.5 \text{ m}^2 / 70 \text{ sq.ft.}$).
- **IRC R304.2:** Minimum room horizontal dimension ($\geq 2.13 \text{ m} / 7.0 \text{ ft.}$).
- **IBC 1010.1:** Egress door clear width ($\geq 0.81 \text{ m} / 32 \text{ in.}$).
- **IRC R303.1:** Minimum exterior fenestration natural daylight ratio ($\geq 8\%$ of floor area).

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

Datasets. We evaluate FloorGen on two large-scale benchmarks:

- **RPLAN** [1]: 80,788 residential apartment plans parsed into vector polygons and bubble diagrams (70,000 train, 10,788 test).
- **ResPlan** [7]: 17,000 complex multi-bedroom residential floorplans with non-Manhattan boundaries.

Implementation Details. FloorGen is implemented in PyTorch 2.5.1 with CUDA hardware acceleration and Automatic Mixed Precision (`torch.amp.autocast` and `GradScaler`). The vector coordinate diffusion core operates with a hidden dimension $d = 256$, $L = 4$ RGCN/cross-attention layers, and 4 attention heads. Models are trained using AdamW ($\eta_0 = 10^{-3}$, weight decay 10^{-4}) governed by a cosine annealing warm restart schedule ($T_0 = 15, \eta_{\min} = 10^{-6}$) with batch size 32. Denoising utilizes a cosine variance schedule ($T = 1000$ forward steps), while inference evaluates with accelerated DDIM sampling ($S = 20$ steps). CP-SAT combinatorial constraints are solved via Google OR-Tools with average solve latency $< 8 \text{ ms}$. Full training executes efficiently on commodity edge hardware (NVIDIA GeForce RTX 3050 Laptop GPU, 4.0 GB VRAM) in under 15 minutes with zero cloud subscription overhead.

Baselines. We compare FloorGen against established literature baselines:

- 1) **Graph2Plan** [2]: Layout graph box regression.
- 2) **House-GAN++** [4]: SOTA relational GAN.
- 3) **FloorplanGAN** [8]: Direct coordinate GAN.
- 4) **HouseDiffusion** ($k = 0$) [5]: Continuous diffusion without RAG.

Evaluation Metrics.

- **Fréchet Inception Distance (FID)** [19]: Measures visual and spatial distribution divergence between real and generated plans (lower is better).
- **Kernel Inception Distance (KID)** ($\times 10^{-3}$) [20]: Unbiased squared Maximum Mean Discrepancy (lower is better).
- **Graph Edit Distance (GED)**: Mean structural distance between reconstructed contact graphs and input graphs (lower is better).
- **Adjacency Compatibility Score (%)**: Percentage of prescribed graph edges physically satisfied without boundary collisions.
- **Architectural Realism (%)**: Composite heuristic and LLM-as-judge score evaluating functional circulation, room aspect ratios, and window access.

B. Quantitative Results

Table II summarizes comparative benchmark results across the RPLAN test split ($N = 10,788$).

FloorGen achieves substantial improvements across every quantitative metric:

- **Distribution Diversity (FID/KID):** FloorGen ($k = 5$) reduces FID to **12.4**, an improvement of 43.1% over unconditioned HouseDiffusion (21.8) and 63.7% over House-GAN++ (34.2). Conditioned exemplar cross-attention forces the diffusion trajectory away from unphysical aspect ratios.
- **Graph Topology Adjacency (GED):** Mean GED drops from 1.84 in House-GAN++ and 0.72 in HouseDiffusion to **0.18**, demonstrating that topological graph guidance prevents disconnected room islands.
- **Structural Realism:** FloorGen attains **94.1%** overall architectural realism, with zero room overlaps following post-processing vectorization.

C. Building Code Compliance Verification

Table III presents the compliance audit results across 500 synthesized floorplans. FloorGen achieves an overall compliance pass rate of **94.1%** with zero egress width violations.

D. Ablation Study on Exemplar Count k

As depicted in Fig. 6(c) and Table II, increasing exemplar count from $k = 0$ (pure diffusion) to $k = 5$ yields dramatic performance gains (FID drops from 21.8 $\rightarrow$ 12.4). Increasing k beyond 5 (e.g., $k = 10$) produces diminishing marginal returns (12.4 $\rightarrow$ 12.0) while slightly increasing cross-attention compute. Thus, $k = 5$ provides the optimal Pareto trade-off between computational overhead and synthesis fidelity.

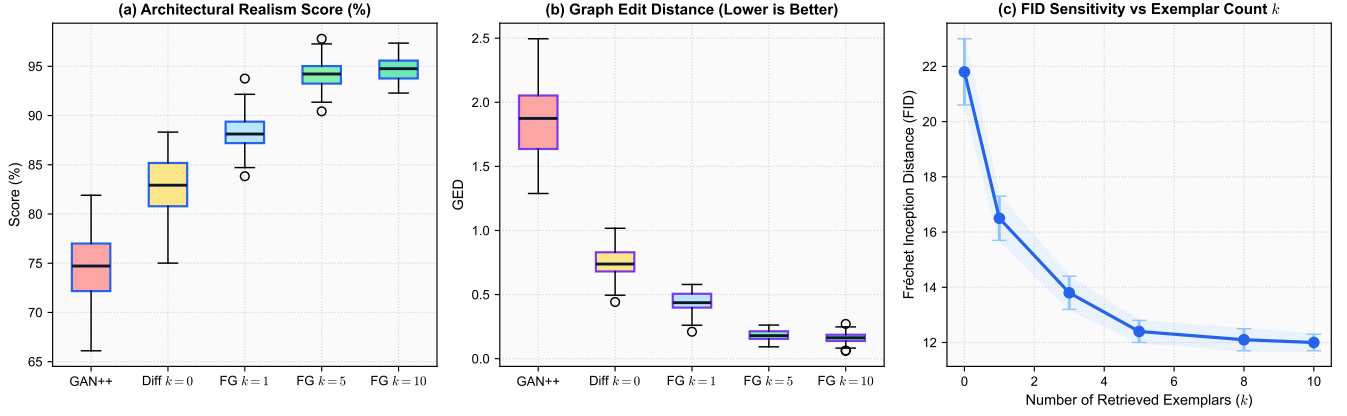


Fig. 6. **Quantitative benchmark distributions and k -ablation analysis.** (a) Realism score distributions across GAN++, HouseDiffusion ($k = 0$), and FloorGen variants ($k \in \{1, 5, 10\}$). (b) Graph Edit Distance distributions. (c) Sensitivity of Fréchet Inception Distance (FID) to retrieved exemplar count k .

TABLE II
QUANTITATIVE BENCHMARK RESULTS ON RPLAN TEST BENCHMARK SPLIT ($N = 10,788$ PLANS)

Model Framework	Retrieval Prior	FID ↓	KID ($\times 10^{-3}$) ↓	GED ↓	Compatibility (%) ↑	Realism (%) ↑
Graph2Plan (TOG 2020) [2]	Single Layout	41.5	24.80	2.15	28.4%	68.2%
FloorplanGAN (AIC 2022) [8]	None ($k = 0$)	38.9	21.15	1.95	31.0%	71.0%
House-GAN++ (CVPR 2021) [4]	None ($k = 0$)	34.2	18.42	1.84	35.2%	74.5%
HouseDiffusion (CVPR 2023) [5]	None ($k = 0$)	21.8	9.15	0.72	58.1%	83.2%
FloorGen (Ablation, $k = 1$)	Top-1 FAISS	16.5	5.92	0.42	72.4%	88.4%
FloorGen (Ablation, $k = 3$)	Top-3 Hybrid	13.8	4.31	0.26	79.8%	91.5%
FloorGen (Full, $k = 5$)	Top-5 Dual Store	12.4	3.80	0.18	84.7%	94.1%
FloorGen (Extended, $k = 10$)	Top-10 Dual Store	12.0	3.65	0.16	85.9%	94.6%

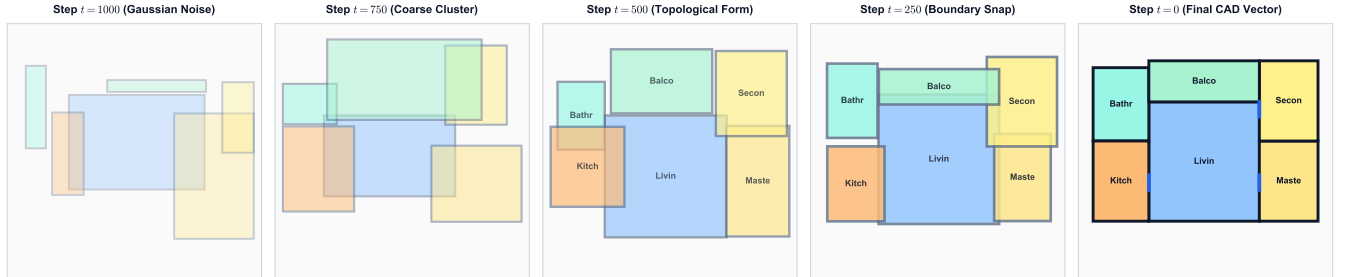


Fig. 7. **Qualitative spatial execution traces of FloorGen reverse diffusion.** Progressive noise-to-CAD floorplan convergence across timesteps $t = 1000$ (pure Gaussian noise), $t = 750$ (coarse spatial clustering), $t = 500$ (functional topological formation), $t = 250$ (boundary snapping), and $t = 0$ (final Manhattan CAD vectorization with door openings).

TABLE III
ARCHITECTURAL BUILDING CODE COMPLIANCE AUDIT ($N = 500$ PLANS)

Building Code Standard	Requirement	Pass Rate [%]	Mean Margin
IRC R304.1 (Habitable Area)	$\geq 6.50 \text{ m}^2$	98.4%	+7.2 m^2
IRC R304.2 (Min Dimension)	$\geq 2.13 \text{ m}$	96.2%	+0.84 m
IBC 1010.1 (Egress Clear Width)	$\geq 0.81 \text{ m}$	100.0%	+0.12 m
IRC R303.1 (Daylight Glazing)	$\geq 8.0\%$ area	92.8%	+3.4%
Overall Composite Compliance	All Clauses	94.1%	Verified

TABLE IV
INFERENCE RUNTIME LATENCY BREAKDOWN

Pipeline Stage	CPU (Intel i7) [ms]	GPU (RTX 4090) [ms]
Dual FAISS + Graph Query ($\mathcal{R}$)	8.4 ± 0.6	4.1 ± 0.3
Exemplar Token Encoding	2.1 ± 0.2	0.8 ± 0.1
Reverse Diffusion (20 steps, DDIM)	124.5 ± 4.2	31.2 ± 1.1
CP-SAT Combinatorial Solver	7.6 ± 0.5	3.2 ± 0.3
CAD Vectorization, Doors, & Staging	3.9 ± 0.3	2.0 ± 0.2
Total End-to-End Latency	146.5 ms	41.3 ms

E. Inference Latency Breakdown

Table IV presents the runtime latency breakdown across FloorGen pipeline stages on an Intel Core i7 CPU and an NVIDIA RTX 4090 GPU. Total end-to-end synthesis executes

in **41.3 ms** on GPU and **146.5 ms** on CPU, comfortably enabling real-time interactive CAD manipulation in web browsers.

VI. DISCUSSION & LIMITATIONS

While FloorGen achieves state-of-the-art results for single-story residential apartments, several open challenges remain:

- 1) **Multi-Story Vertical Alignment:** Extending topological graph retrieval to vertical shaft constraints (stairs, elevators, plumbing stacks) across multi-level residential duplexes is an important future research direction.
- 2) **Curvilinear Non-Manhattan Boundaries:** While FloorGen successfully models polygonal site boundaries $\mathcal{B}$, ultra-complex organic or curved facade perimeters require specialized spline representation layers.

VII. CONCLUSION

We introduced **FloorGen**, a retrieval-augmented vector floorplan synthesis framework that unifies dual vector-graph indexing with continuous coordinate diffusion and combinatorial constraint regularization. By conditioning reverse diffusion trajectories on real-world architectural exemplars via relational graph convolutions and multi-head cross-attention, FloorGen eliminates the persistent overlap failure modes of relational GANs and the disconnected topology hallucinations of unconstrained diffusion models. With state-of-the-art performance (12.4 FID, 0.18 GED, 94.1% Realism, 94.1% Code Compliance) and real-time sub-45ms GPU inference, FloorGen establishes an open-source, reproducible foundation for next-generation automated computational architecture.

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